

Permutation Invariance and Paradox

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1 Introduction

Consider a welfare distribution in world 1 that ascribes 1 utility to Alice, 2 utility to Bob, and 3 utility to Carlos, and a welfare distribution in world 2 that ascribes 2 utility to Alice, 3 utility to Bob, and 1 utility to Carlos. Intuitively, we think that the overall value of world 1 is equivalent to the overall value of world 2. What could motivate the intuition of the moral irrelevance of individual identities? Here, proponents of impartiality appeal to the principle of Permutation Invariance—that the ethical comparison of welfare distributions should be unchanged by permutations on the set of individuals, even if there are infinite permutations on an infinite population¹. To secure impartiality, we should not grant additional moral weight to any specific individual, and thus we ought to be indifferent between welfare distributions that only change the identity of individuals ascribed to some welfare level in the welfare distribution. Indeed, it seems that if we are to deny permutation invariance, then we would be led to the conclusion that the overall value of some world would discriminate between disjoint sets of individuals of the same cardinality. How one would make sense of this is at least *prima facie* mysterious.

But permutation invariance is not without problems. Consider the following setup²:

Let I be an infinite set of individuals, $\mathbb{W}$ a set of welfare levels, some of which are greater than others, and $w : I \rightarrow \mathbb{W}$ a welfare distribution, a function that maps a welfare level in $\mathbb{W}$ to an individual in I . Here, we are interested in the at least as good as relation $\succeq$ on the set $\mathbb{W}^I$ of all welfare distributions, with strict betterness ($\succ$) is defined as the asymmetric part of $\succeq$, equal goodness ($\sim$) its symmetric

¹Given completeness and transitivity, Goodman (2025) has shown that permutation invariance is equivalent to the otherwise stronger claim of Anonymity: $w \sim \pi w$ for every welfare distribution w and permutation on the set of individuals π . For my paper, I assume completeness and transitivity hold, and thus Permutation Invariance is indistinguishable from Anonymity; I will therefore use these terms interchangeably.

²This setup formulation can be found in Nebel (forthcoming)

part. Now, for any permutation π on the set of individuals and any welfare distribution w , let πw denote the welfare distribution that assigns welfare level $w(\pi(i))$ to each individual $i \in I$. According to

Permutation Invariance: For any distributions w and v , and any permutation $\pi : I \rightarrow I$, $w \succeq v$ if and only if $\pi w \succeq \pi v$.

The standard argument against Permutation Invariance appeals to the Strong Pareto principle that what is better for some and at least as good for all is better overall:

Strong Pareto: For any distributions w and v , if there is some individual i such that $w_i \succ v_i$ and $w_j \succeq v_j$ for every other individual j , then $w \succ v$.

From here, the argument is straightforward: Imagine an infinite population with individuals ascribed to either a high welfare state, or a low welfare state. If we improve the condition of one individual from a low welfare state to a high welfare state, Strong Pareto implies that the overall value of this new population is strictly better than the original population, since one individual was benefited and no one else was harmed. However, because both populations still contain an infinite number of people in the low welfare state and an infinite number of people in the high welfare state, Permutation Invariance implies that the overall value of the original infinite population is equally as good as the overall value of the new infinite population, since the different welfare distributions between populations are mere permutations of one another. Thus, we arrive at a contradiction: Strong Pareto tells us the new population is strictly better than the original population, while Permutation Invariance tells both populations are equally as good.³ Other arguments against permutation invariance have been made by Jonsson and Peterson (2020), Russell (2025), Nebel (2025), and Nebel (Forthcoming).

However, in all these cases, the challenge to permutation invariance in the infinite population case relies on a purported incompatibility with a decision-theoretic dominance principle of some sort. I intend to show the untenability of permutation invariance without any sort of these principles. I do this by showing that permutation invariance, along with the standard Kolmogorov probability axioms and expected utility reasoning, give rise to inconsistency via an analogous line of reasoning to the Banach-Tarski Paradox. The upshot of this argument is that there cannot exist a coherent probability measure over the set of permutations on a set of individuals, and therefore our ability to discriminate between lotteries via expected utility reasoning collapses.

³This argument was initially made by Van Liedekerke (1995)

2 Main Result

Assume an infinite population whose identities are mapped to the natural numbers, $\mathbb{N} = \{1, 2, 3, \dots\}$. Let S_∞ be the group of all permutations of $\mathbb{N}$, representing the state space of all possible identity assignments. To evaluate risk under a veil of ignorance, we assume an impartial observer is rational according to standard decision theory: they satisfy the von Neumann-Morgenstern axioms, possessing a state-independent utility function over welfare distributions, and they possess complete preferences over the state space. The principle of Permutation Invariance dictates that our observer considers the identities of individuals morally arbitrary; therefore, they must be strictly indifferent between facing a lottery over a subset of identity assignments X and a lottery over that same subset shifted by any permutation g .

Principles. To derive the proof, we require five principles. The first three define a standard, coherent probability measure. The fourth defines a modification to the original Permutation Invariance principle. The fifth is a structural fact from combinatorial group theory.

Principle 1: Total Probability (TP) The probability measure of the entire group must equal 1. $\mu(G) = 1$.

Principle 2: Non-Negativity (NN) For any subset $X \subseteq G$, its probability must be non-negative. $\mu(X) \geq 0$.

Principle 3: Finite Additivity (FA) If subsets A and B are disjoint ($A \cap B = \emptyset$), the probability of their union is the sum of their individual probabilities. $\mu(A \cup B) = \mu(A) + \mu(B)$.

Principle 4: Measure Permutation Invariance (MPI) The ethical evaluation is anonymous. Shifting a subset of permutations X by multiplying it by any permutation g does not change its probability mass. $\mu(gX) = \mu(X)$.

Principle 5: The Free Group Embedding (FGE) The infinite symmetric group S_∞ contains a subgroup isomorphic to F_2 , the free group on two generators (let us call these generators a and b). If S_∞ admits a probability measure satisfying TP, NN, FA, and MPI, then any subgroup within it, including F_2 , must also admit such a measure.⁴

Although MPI does not follow immediately from PI, its derivation is fairly straightforward given Expected Utility Theory. The proof is as follows:

⁴This was proven by von Neumann (1929). For our purposes, we need only the result that the property of amenability is inherited by subgroups, and thus any result that shows we cannot have a coherent probability measure over F_2 also shows we cannot have a coherent probability measure over S_∞ .

- (P1) Standard PI states: $w \succeq v \iff \pi w \succeq \pi v$. If we set v equal to w , the principle trivially yields $w \sim \pi w$.
- (P2) Let $X \subseteq S_\infty$ be a subset of welfare permutations on identities. Let $g \in S_\infty$ be an arbitrary structural shift, creating the shifted subset gX . Let w^* be a highly desirable welfare distribution and w_0 be a terrible one so that $w^* \succ w_0$. Construct two lotteries:
- Lottery L: Yields w^* if the realized identity state falls in subset X and w_0 otherwise.
 - Lottery gL: Yields w^* if the realized identity state falls in subset gX and w_0 otherwise.
- (P3) Because the observer considers identities ethically arbitrary (from Step 1), and because the payoffs are identical, they must be strictly indifferent between playing Lottery L and playing Lottery gL. Therefore, $L \sim gL$.
- (P4) Because $L \sim gL$, EUT demands that their expected utilities are strictly equal:

$$\mathbb{E}[U(L)] = \mathbb{E}[U(gL)]$$

Let μ be the observer's subjective probability measure over the state space of permutations S_∞ . We write out the expected utility for both lotteries:

$$\mathbb{E}[U(L)] = \mu(X) \cdot U(w^*) + (1 - \mu(X)) \cdot U(w_0)$$

$$\mathbb{E}[U(gL)] = \mu(gX) \cdot U(w^*) + (1 - \mu(gX)) \cdot U(w_0)$$

- (P5) Via algebraic manipulation, we can equate the two expected utilities:

$$\mu(X)U(w^*) + (1 - \mu(X))U(w_0) = \mu(gX)U(w^*) + (1 - \mu(gX))U(w_0)$$

$$\mu(X)[U(w^*) - U(w_0)] + U(w_0) = \mu(gX)[U(w^*) - U(w_0)] + U(w_0)$$

Subtracting $U(w_0)$ from both sides:

$$\mu(X)[U(w^*) - U(w_0)] = \mu(gX)[U(w^*) - U(w_0)]$$

Because w^* is strictly preferred to w_0 , we know $[U(w^*) - U(w_0)] > 0$. We divide both sides by this positive constant, yielding:

$$\mu(X) = \mu(gX)$$

MPI is thus a consequence of accepting PI and Expected Utility Theory.

With MPI established and TP, NN, FA, and FGE taken to hold, we can make the inconsistency argument as follows:

Proof. Let F_2 be the free group generated by a and b . Let $W(x)$ denote the set of all "words" (elements of the group) that begin with the letter x , where $x \in \{a, a^{-1}, b, b^{-1}\}$. Let e denote the empty word (the identity element)

- (P1) Every word in F_2 either is the empty word e , or it begins with exactly one of the four letters: a, a^{-1}, b , or b^{-1} . (By definition of a free group).
- (P2) Therefore, F_2 can be partitioned into exactly five mutually exclusive and collectively exhaustive subsets: $F_2 = \{e\} \cup W(a) \cup W(a^{-1}) \cup W(b) \cup W(b^{-1})$. (From P1).
- (P3) $\mu(F_2) = \mu(\{e\}) + \mu(W(a)) + \mu(W(a^{-1})) + \mu(W(b)) + \mu(W(b^{-1}))$. (From P2 and FA).
- (P4) $\mu(F_2) = 1$. (From TP).
- (P5) If we take the subset $W(a^{-1})$ and multiply every element within it by the generator a , the a and a^{-1} cancel out. This algebraic shift generates every possible word in the entire group except words that strictly begin with a . Thus, $aW(a^{-1}) = F_2 \setminus W(a)$. (By the algebraic properties of F_2).
- (P6) $\mu(aW(a^{-1})) = \mu(F_2 \setminus W(a))$. (Applying the measure μ to both sides of P5).
- (P7) $\mu(aW(a^{-1})) = \mu(W(a^{-1}))$. (From MPI: multiplying by a does not change the measure).
- (P8) $\mu(F_2 \setminus W(a)) = \mu(F_2) - \mu(W(a)) = 1 - \mu(W(a))$. (From FA applied to complementary sets, and P4).
- (P9) $\mu(W(a^{-1})) = 1 - \mu(W(a))$. (Substituting P7 and P8 into P6).
- (P10) $\mu(W(a)) + \mu(W(a^{-1})) = 1$. (Rearranging P9).
- (P11) By identical algebraic logic applied to the generator b , shifting $W(b^{-1})$ by b yields $\mu(W(b)) + \mu(W(b^{-1})) = 1$. (Parallel deduction to P5-P10).
- (P12) $\mu(W(a)) + \mu(W(a^{-1})) + \mu(W(b)) + \mu(W(b^{-1})) = 2$. (Adding P10 and P11).
- (P13) Substituting P12 back into our total partition equation (P3) yields:
 $\mu(F_2) = \mu(\{e\}) + 2$.

- (C1) Because $\mu(\{e\}) \geq 0$ (from NN), P13 requires that $\mu(F_2) \geq 2$. However, P4 requires that $\mu(F_2) = 1$. Therefore, $1 \geq 2$, which is a strict mathematical contradiction.
- (C2) Therefore, it is impossible for any measure μ to satisfy TP, NN, FA, and MPI simultaneously over F_2 . Because F_2 is a subgroup of S_∞ (from FGE), S_∞ is non-amenable. Measure Permutation Invariance across an infinite population is mathematically incompatible with the existence of a coherent expected utility evaluation.

3 Conclusion

Since I find the principles invoked in my above result to be mathematically indispensable, I take this to be a strong reason to reject Permutation Invariance. At the very least, we're forced into a dilemma: abandon the axioms of expected utility theory, or reject Permutation Invariance in the infinite setting. Given the philosophical costs of rejecting any one EUT axiom,⁵ the most tenable resolution is to concede that Permutation Invariance does not scale to the infinite level. While ethical indifference to permutations of a welfare distribution over a set of finite individuals remains both mathematically and philosophically sound, demanding strict permutation invariance across an infinite horizon demands a structural symmetry that the mathematics of rational choice cannot accommodate.

References

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